Bartz Nozzle Wall-Temperature Solver

Matthew Chu — Probation task — 12 October 2026

1. Program and governing model

Deliverable. `bartz_nozzle.py` accepts nozzle geometry and operating conditions and returns inner/outer wall temperature along the nozzle, the gas-side heat-transfer coefficient and inner heat flux. It supports steady and finite-duration transient heating. The example is synthetic; no team hardware data or thermal measurements were supplied.

Inputs, in SI units. The geometry CSV contains `x_m`, `r_inner_m` and `wall_thickness_m`; thickness is radial. Include a unique interior minimum-radius throat and both converging and diverging sections. The JSON specifies absolute chamber pressure p_0, stagnation temperature T_0, γ, c_p, dynamic viscosity μ, Prandtl number Pr, characteristic velocity c^* and axial throat-contour curvature radius r_c. It also supplies wall k_w, ρ_w, c_w, initial temperature, external film coefficient h_o, fluid/surroundings temperatures T_b, T_s, emissivity ϵ_o, burn duration and numerical settings. r_c is distinct from the throat flow radius.

Gas and Bartz calculation. With $A = \pi r_i^2$, $A_t = \min A$ and $D_t = 2\sqrt{A_t/\pi}$, bracketed area–Mach roots select subsonic flow before the throat, $M = 1$ at the throat and supersonic flow downstream [3]. Set $\chi = 1+(\gamma-1)M^2/2$, $T_g = T_0/\chi$ and $p_g = p_0\chi^{-\gamma/(\gamma-1)}$. The implemented coherent-SI Bartz correlation is [1,2]

$$h_g = \frac{0.026}{D_t^{0.2}} \frac{\mu^{0.2} c_p}{\mathrm{Pr}^{0.6}} \left(\frac{p_0}{c^*}\right)^{0.8} \left(\frac{D_t}{r_c}\right)^{0.1} \left(\frac{A_t}{A}\right)^{0.9} \sigma, \qquad \sigma = \left[\tfrac{1}{2}(T_{w,g}/T_0)\chi + \tfrac{1}{2}\right]^{-0.68} \chi^{-0.12}. \tag{1}$$

The supplied gas properties use chamber-reference values. There is no additional g_0 conversion factor in this SI form. The turbulent recovery-temperature driving potential is [2]

$$T_{aw} = T_g \left[1 + \mathrm{Pr}^{1/3} \tfrac{\gamma-1}{2} M^2\right], \qquad q_g'' = h_g(T_{w,g})(T_{aw} - T_{w,g}). \tag{2}$$

Thus T_{aw} alone is not a prediction of cooled wall temperature; the wall boundary problem must also be solved.

Steady wall. At each axial station the wall is a local cylindrical ring, $r_o = r_i + t$, with heat rates per axial metre:

$$\begin{aligned} Q_g' &= 2\pi r_i h_g(T_{w,g})(T_{aw} - T_{w,g}), \qquad Q_k' = \frac{2\pi k_w (T_{w,g} - T_{w,o})}{\ln(r_o/r_i)}, \\ Q_o' &= 2\pi r_o \left[h_o(T_{w,o} - T_b) + \epsilon_o \sigma_{SB}(T_{w,o}^4 - T_s^4)\right]. \end{aligned} \tag{3}$$

The code enforces $Q_g' = Q_k' = Q_o'$. It substitutes $T_{w,g} = T_{w,o} + \ln(r_o/r_i)Q_o'/(2\pi k_w)$ into the gas balance and uses a bracketed Brent solve for $T_{w,o}$, updating $h_g(T_{w,g})$ during every residual evaluation.

Transient wall. Conservative radial finite volumes discretize $\rho_w c_w \partial_t T = (k_w/r)\partial_r(r\partial_r T)$. Cell capacity is $C_j' = \rho_w c_w \pi(r_{j+1/2}^2 - r_{j-1/2}^2)$; adjacent-centre conductance is $G_{j,j+1}' = 2\pi k_w / \ln(r_{j+1}/r_j)$. Nonlinear massless surface temperatures couple each film to the boundary half-cell. SciPy BDF integration uses a sparse Jacobian pattern [4]. An extra state integrates $Q_g' - Q_o'$ for comparison with $\sum_j C_j'(T_j - T_i)$.

Scope. Choked isentropic gas, turbulent empirical Bartz heat transfer, constant properties, independent radial rings and constant operating conditions during the burn. Axial conduction, shocks/separation, gas radiation, ablation and coolant-channel flow are omitted. Cooling is a prescribed boundary condition. At an instantaneous gas-temperature step, the discrete massless surface can jump while cell temperatures remain at their initial value; radial refinement reduces this half-cell effect.

2. Reproduction and example results

Run from the submission folder with Python 3.10 or later. The solver requires NumPy 2.0+, SciPy 1.13+ and Matplotlib 3.8+; the verification driver also uses standard-library `unittest`.

```
python -m pip install "numpy>=2.0" "scipy>=1.13" "matplotlib>=3.8"
cd "Example, Verify and Brief Report"
python ../bartz_nozzle.py --geometry nozzle_geometry.csv --config operating_conditions.json --output . --mode both
python test_bartz.py
```

Each displayed command is one continuous command line. `--mode steady` and `--mode transient` select one solution. Input validation rejects invalid geometry/properties and unchoked back-pressure conditions; possible internal shocks are flagged, not resolved.

Submitted outputs. `wall_temperature_profile.csv` stores x, Mach number, gas conditions, steady/burn-end wall temperatures, h_g and q''_g. `transient_surface_temperatures.csv` stores the complete surface histories. `summary.json` records extrema and the energy audit. `unit_test_results.json` and `verification.json` record test outcomes and convergence studies. Figures are embedded here; duplicate plot files and internal radial-field archives are omitted from the submission. Running the solver regenerates those additional outputs.

Synthetic input case. Length 200 mm, throat at $x = 80$ mm, $D_t = 24$ mm, $r_c = 25$ mm, radial wall thickness 3 mm and 121 axial stations. The material and cooling values are illustrative, not a specified alloy or team cooling system.

Parameter	Value	Parameter	Value
p_0	2.00 MPa absolute	T_0	2800 K
$\gamma,\ c_p$	1.20; 2400 J/(kg K)	$\mu,$ Pr	8.0×10^{-5} Pa s; 0.70
c^*	1631.84 m/s	Back pressure	0 Pa
$k_w,\ \rho_w,\ c_w$	30 W/(m K); 7800 kg/m^3; 600 J/(kg K)	T_i, T_b, T_s	293.15 K
$h_o,\ \epsilon_o$	8000 W/(m^2 K); 0.70	Heating time	1.00 s
Radial cells	24	BDF rtol, atol	10^{-6}; 10^{-5} K

Calculated quantity	Result
Mass flow $p_0 A_t/c^*$; exit Mach number	0.55445 kg/s; 3.1853
Peak inner wall: steady / one-second heating	1786.54 K at 78 mm / 1476.95 K at 80 mm
Peak outer wall: steady / one-second heating	1079.31 K / 758.50 K
Peak steady h_g; peak steady inner heat flux	8.025 kW/(m^2 K); 7.918 MW/m^2

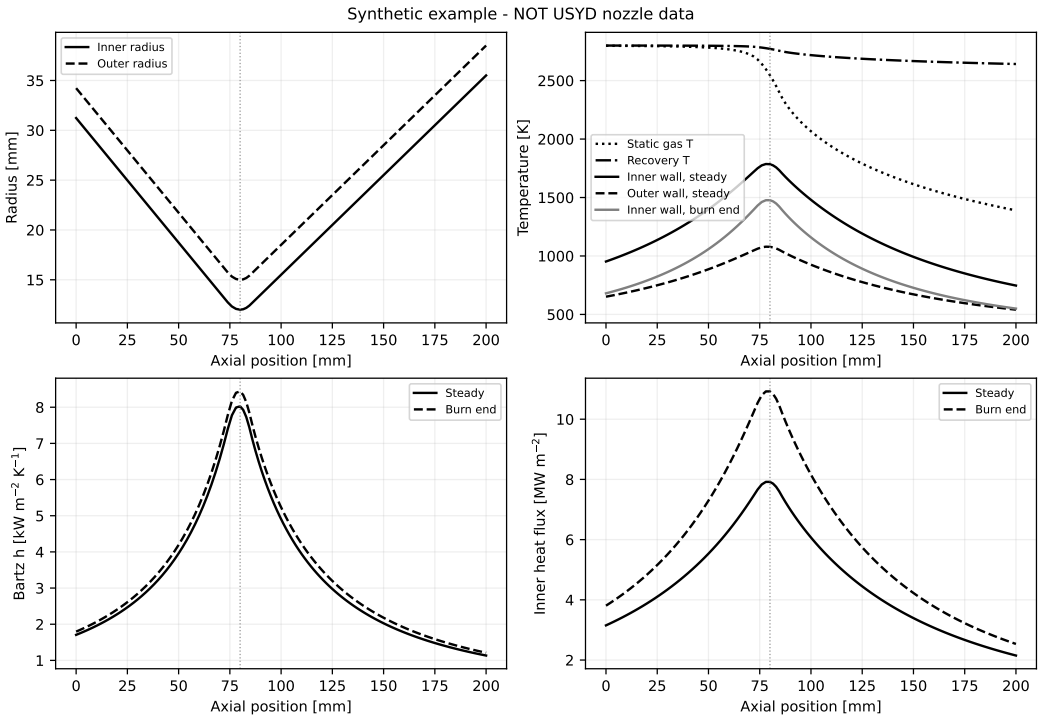


Figure 1. Synthetic axial results; the dotted vertical line marks the throat. Thermal storage keeps the one-second wall temperature below its steady value.

3. Verification and references

Reproducible checks. `test_bartz.py` is self-contained apart from the solver and its scientific Python dependencies. It runs 14 automated tests and then independent radial, axial and time-tolerance studies. All 14 tests passed, with zero failures or errors.

Check	Evidence and acceptance criterion
Gas solution	Known $\gamma = 1.4$, $M = 2$, $A/A_t = 1.6875$ recovered; correct subsonic/supersonic branches and stagnation relations.
Bartz feedback	Fixed-wall $p_0^{0.8}$, $\mu^{0.2}$, $(c^*)^{-0.8}$ and $r_c^{-0.1}$ scalings; analytic heat-rate derivative agrees with a centred difference to relative tolerance 10^{-8}.
Steady conservation	Gas/conduction/outer heat rates agree; synthetic full-case maximum relative mismatch 2.8×10^{-14}. Insulated limit gives $T_{w,g} = T_{w,o} = T_{aw}$ and zero heat transfer; radiation-only boundary also balances.
Radial discretization	Exact logarithmic steady wall profile inserted into the finite-volume equations gives maximum cell $\lvert dT/dt \rvert < 2 \times 10^{-5}$ K/s.
Transient solution	Internal fluxes cancel; temperatures remain bounded. A 20-s test converges to the independently computed steady surfaces within 0.05 K. Full example stored-energy/net-heat mismatch is 1.7×10^{-15} relative.
Input/boundary checks	Invalid geometry and properties rejected; unchoked back pressure rejected; spatially varying external cooling profiles accepted.

The energy audit integrates net boundary heat as an extra ODE state. Its small mismatch demonstrates conservation of the discrete system; it is not an independent estimate of integration error or empirical prediction accuracy.

Numerical refinement

The radial study evaluates the throat at 1 s with rtol $= 2 \times 10^{-7}$ and atol $= 2 \times 10^{-6}$ K. Each axial ring is independent, so throat refinement does not require a full axial transient.

Radial cells	Throat inner wall [K]	Throat outer wall [K]
12	1476.82072	758.39174
24	1476.95357	758.50434
48	1476.98705	758.53281

The 24-to-48-cell inner-wall difference is 0.03348 K. Tightening time tolerances from $10^{-5}/10^{-4}$ to $2 \times 10^{-7}/2 \times 10^{-6}$ changes the maximum sampled final inner-wall temperature by 0.00327 K. For 121, 241 and 481 axial stations, steady sampled peaks are 1786.5397, 1787.3412 and 1787.3498 K; the final refinement changes the peak by 0.00860 K.

Interpretation. These checks establish consistency and convergence within the implemented model. Experimental validation remains outstanding. For the actual nozzle, replace geometry, gas/material properties, burn duration and cooling inputs, then compare wall temperature or heat flux against measurements or an appropriate higher-fidelity model. The synthetic results do not establish a permissible hardware operating temperature.

References

[1] D. R. Bartz, "A Simple Equation for Rapid Estimation of Rocket Nozzle Convective Heat Transfer Coefficients," *Journal of Jet Propulsion*, 27(1), 49–51, 1957. https://doi.org/10.2514/8.12572.

[2] R. M. Traci, J. L. Farr Jr., T. Laganelli and J. Walker, *A Thermal Management Systems Model for the NASA GTX RBCC Concept*, NASA/CR-2002-211587, 2002, printed pp. 6–7 (recovery enthalpy and Bartz equations). https://ntrs.nasa.gov/citations/20020065525.

[3] NASA Glenn Research Center, *Isentropic Flow Equations*, updated 13 May 2021, equations 6–9. https://www.grc.nasa.gov/www/k-12/airplane/isentrop.html.

[4] SciPy, *scipy.integrate.solve_ivp*, official API documentation: BDF method and `jac_sparsity`. https://docs.scipy.org/doc/scipy/reference/generated/scipy.integrate.solve_ivp.html.